

journey is in the continuous improvement mode. Most of these tools have built-in rules, but they can be tweaked or extended with custom rules if required.

Challenges

Though most of these tools are rich in features, their real benefit can be perceived only if the tool is properly understood and configured; otherwise, there will be a risk of missing on leveraging the full potential. This challenge can be mitigated by having a dedicated team working on the features of these tools, with a constant eye on timely feature upgrades. Also, the same learning has to be propagated to the developer community on a regular basis to keep them updated about the changes.

Final comments

Overall, each tool has its own strong points based on the respective individual features. However, SonarQube has an edge over the others with its comprehensive feature list that covers support for multiple programming languages, access to a centralised repository and maintenance, duplicate code

identification, easy integration with most of the development and build tools, and many more. Last but not the least, its rich dashboard feature makes it an obvious choice for both the developer community and the organisation. However, more than the tools, it is the discipline which has to be inculcated and maintained in the teams that makes the entire code review process effective. The tailoring of the processes can be done based on the individual situations, but should be arrived at after taking all the stakeholders into confidence.

 **Note:** Views expressed by the author are his own and presented here in his personal capacity. These are in no way representative and/or reflective of the organisation he is associated with.

END 

 **By: Sharath T.**

The author has 21+ years of experience and an established track record in successful product development, strategy, delivery management and handling of cross-functional responsibilities. Currently he is director, product development at Oracle.

Continued From Page...94

proposal called the transient field can be used to pass private data from the client (or data that chaincode will use to generate private data) to chaincode invocation on the peer. The chaincode can retrieve the transient field by calling the *GetTransient()* API. This transient field gets excluded from the channel transaction.

Access control for private data

Until version 1.3, access control to private data based on collection membership was enforced for peers only. Access control based on the organisation of the chaincode proposal submitter was required to be encoded in chaincode logic.

Querying private data

Private collection data can be queried just like normal channel data, using shim APIs:

```
GetPrivateDataByRange(collection, startKey, endKey string)
GetPrivateDataByPartialCompositeKey(collection, objectType
string, keys []string)
```

And for the CouchDB state database, JSON content queries can be passed using the shim API:

```
GetPrivateDataQueryResult(collection, query string)
```

Upgrading a collection definition

If a collection is referenced by a chaincode, the chaincode will use the prior collection definition unless a new collection

definition is specified at upgrade time. If a collection configuration is specified during the upgrade, a definition for each of the existing collections must be included, and you can add new collection definitions.

Collection updates become effective when a peer commits the block that contains the chaincode upgrade transaction.

Updates related to private data collections in Hyperledger Fabric V1.4 release

Querying updates: A collection configuration option *memberOnlyRead* can automatically enforce access control based on the organisation of the chaincode proposal submitter.

Reconciliation updates: A background process allows peers who are part of a collection to receive data they were entitled to receive but did not yet receive — because of a network failure, for example — by keeping track of private data that was ‘missing’ at the time of block commit. The peer will periodically attempt to fetch the private data from other collection member peers that are expected to have it. **END** 

 **By: Swapnil Kulkarni**

The author is an open source enthusiast with experience in blockchain, cloud native solutions, containers and enterprise software product architectures. He has diverse experience with different private, hybrid cloud architectures. He is a technology hobbyist and writes at *cloudnativetech.wordpress.com*.